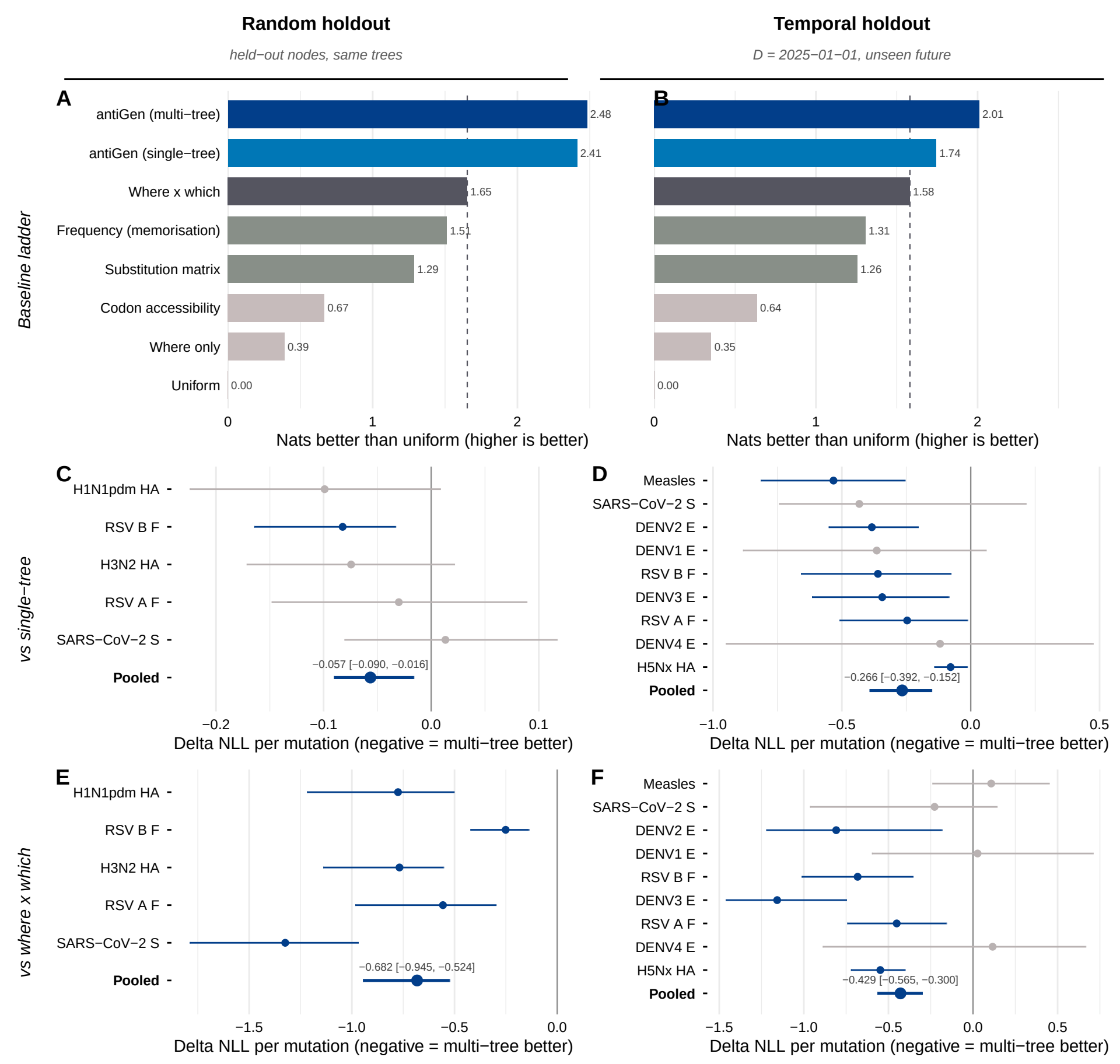
Random holdout
held−out nodes, same trees
Temporal holdout
D = 2025−01−01, unseen future
Baseline ladder
A
antiGen (multi−tree)
antiGen (single−tree)
Where x which
Frequency (memorisation)
Substitution matrix
Codon accessibility
Where only
Uniform
2.48
2.41
1.65
1.51
1.29
0.67
0.39
0.00
0
1
2
Nats better than uniform (higher is better)
B
2.01
1.74
1.58
1.31
1.26
0.64
0.35
0.00
0
1
2
Nats better than uniform (higher is better)
vs single−tree
C
H1N1pdm HA
RSV B F
H3N2 HA
RSV A F
SARS−CoV−2 S
Pooled
−0.057 [−0.090, −0.016]
−0.2
−0.1
0.0
0.1
Delta NLL per mutation (negative = multi−tree better)
D
Measles
SARS−CoV−2 S
DENV2 E
DENV1 E
RSV B F
DENV3 E
RSV A F
DENV4 E
H5Nx HA
Pooled
−0.266 [−0.392, −0.152]
−1.0
−0.5
0.0
0.5
Delta NLL per mutation (negative = multi−tree better)
vs where x which
E
H1N1pdm HA
RSV B F
H3N2 HA
RSV A F
SARS−CoV−2 S
Pooled
−0.682 [−0.945, −0.524]
−1.5
−1.0
−0.5
0.0
Delta NLL per mutation (negative = multi−tree better)
F
Measles
SARS−CoV−2 S
DENV2 E
DENV1 E
RSV B F
DENV3 E
RSV A F
DENV4 E
H5Nx HA
Pooled
−0.429 [−0.565, −0.300]
−1.5
−1.0
−0.5
0.0
0.5
Delta NLL per mutation (negative = multi−tree better)